

原始提示詞：

請寫出有關 *Zeeman Effect* 的報告

使用技巧的提示詞：

你是物理實驗課的老師，請寫出一篇 *Zeeman Effect* 的報告，使得學生能夠透過你的報告了解實驗相關原理與方法。請分成以下段落：

I. Introduction (可能提到背景、介紹、原理、應用、實驗目標等)

II. Experimental Method

III. Results and Discussion

IV. Conclusion

V. Appendix(非必要)

最後是 *References*(若有)

此段提示詞一次使用到三個技巧：

1. System Prompting：Defines the model's fundamental capabilities and overarching purpose.
此技巧使得 LLM 知道所需段落、結構與我想要的內容重點。(如 Zeeman 的生平或歷史就不是我要的重點)
2. Role prompt：Frames the model's output style and voice. It adds a layer of specificity and personality.
此技巧使得 LLM 知道目標是教會學生，文章需有教學性。
3. Contextual prompt：Provides immediate, task-specific information to guide the response. It's highly specific to the current task or input, which is dynamic.
此技巧使得 LLM 知道更具體的情境，是老師要教學生，而不是物理學家發表論文。

References: Prompt Engineering by Lee Boonstra

回覆比較：

原始提示詞：



Zeeman Effect Report (塞曼效應報告)

一、實驗目的 (Purpose)

1. 了解塞曼效應 (Zeeman Effect) 的物理原理。

2. 觀察原子能階在外加磁場作用下的分裂現象。

3. 測量譜線分裂的程度，以求出電子的磁矩 (magnetic moment) 或驗證玻爾磁子 (Bohr magneton) 的值。

二、實驗原理 (Theory)

1. 現象概述

塞曼效應是指當原子或離子在外加磁場中發光時，其光譜線發生分裂的現象。

此效應最早由荷蘭物理學家 Pieter Zeeman 於 1896 年發現，並由 Lorentz 成功解釋，因此又稱為洛倫茲-塞曼效應。

2. 理論基礎

在沒有外加磁場時，原子的能階僅由電子的量子數 n, l, s, j 決定。

當加上磁場 B 時，能階會因磁矩與磁場交互作用而發生微小分裂：

$$\Delta E = m_j g_J \mu_B B$$

使用技巧的提示詞：

I. Introduction

The **Zeeman Effect** refers to the phenomenon in which the spectral lines of atoms or ions are split into multiple components when subjected to an external magnetic field.

This effect was first observed by **Pieter Zeeman** in 1896 and theoretically explained by **Hendrik Lorentz**, both of whom were later awarded the Nobel Prize in Physics in 1902.

1. Physical Background and Principle

In the absence of a magnetic field, atomic energy levels depend only on the quantum numbers n, l, s , and j .

When a magnetic field B is applied, the **magnetic dipole moment** associated with the electron's angular momentum interacts with the field, leading to a small shift in the energy levels:

$$\Delta E = m_j g_J \mu_B B$$

where

- m_j is the magnetic quantum number,
- g_J is the Landé g-factor,
- $\mu_B = \frac{e\hbar}{2m_e}$ is the **Bohr magneton**,
- B is the magnetic field strength.

This energy difference produces multiple transitions between split levels, resulting in the **splitting of spectral lines**.

2. Types of Zeeman Effect

There are two major types:

- **Normal Zeeman Effect:** occurs when the total electron spin $S = 0$. Each spectral line splits into **three components** ($\Delta m = 0, \pm 1$).
- **Anomalous Zeeman Effect:** occurs when $S \neq 0$, and the pattern of splitting is more complicated because of spin-orbit coupling.

3. Polarization Properties

Each of the split lines exhibits distinct polarization:

- **π -component ($\Delta m = 0$)** → linearly polarized, parallel to the magnetic field.
 - **σ^+ and σ^- components ($\Delta m = \pm 1$)** → circularly polarized in opposite directions.
- When viewed **perpendicular** to the magnetic field, both π and σ lines are visible; along the field direction, only σ lines can be seen.

4. Applications

The Zeeman Effect is widely applied in:

- **Astrophysics** – to measure magnetic fields of stars and sunspots.
- **Laser and atomic spectroscopy** – for fine structure and energy level analysis.
- **Magneto-optical instruments** – such as Zeeman-based wavelength calibration devices.

5. Objective

The aim of this experiment is to **observe the Zeeman splitting** of spectral lines in a mercury or cadmium discharge lamp under a magnetic field, and to **verify the linear relationship** between the splitting and the field strength.

From the data, we can also **estimate the value of μ_B** Bohr magneton and compare it with the theoretical value.

II. Experimental Method

1. Apparatus

- Discharge lamp (e.g., Cadmium or Mercury)
- Electromagnet with adjustable current supply
- Hall probe and gaussmeter (for measuring magnetic field strength)
- Fabry–Perot interferometer or high-resolution spectrometer
- Polarizer and optical filters
- CCD camera or photodetector for capturing spectra
- Optical bench and alignment system

2. Procedure

1. Setup:

Mount the discharge lamp between the poles of the electromagnet so that the emitted light travels either parallel or perpendicular to the magnetic field.

2. Magnetic Field Calibration:

Vary the current in the electromagnet and measure the magnetic field B using a Hall probe.

3. Observation of Spectral Lines:

Use a Fabry–Perot interferometer to observe interference rings of a chosen spectral line (e.g., the red line of Cd at 643.8 nm).

4. Record Data:

Gradually increase the magnetic field and capture the pattern of split interference fringes.

5. Measurement:

Determine the wavelength separation $\Delta\lambda$ from the spacing of interference rings.

6. Analysis:

Plot $\Delta\lambda$ versus B , and from the slope calculate the Bohr magneton using the relationship:

$$\mu_B = \frac{hc\Delta\lambda}{B\lambda^2\Delta m}$$

使用技巧的提示詞產生的結果比較有論文風格，並同時具有教學性，除了教實驗相關知識，也為”怎麼寫實驗報告”做了一個很好的範例。